THE HOM GROUP OUT OF THE ALTERNATING POWER OF THE EXTENSION BY ZERO IS THE ANTI-INVARIANT PART OF THE SECTIONS OVER THE CONFIGURATION SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The diagonal of a separated etale morphism is an open and closed immersion. Morphisms of Spaces, labels spaces-morphisms-lemma-etale-unramified and spaces-morphisms-lemma-diagonal-unramified-morphism (invoked at proof lines 1088 and 1090). They show that the complement W_p of the diagonals in the $(p+1)$-fold fibre product is an open and closed subspace, so that the sections over it split off.
- The tensor power of the extension by zero splits along the complement of the diagonals. Same chapter spaces-cohomology.tex, label lemma-product-is-tensor-product (invoked at proof line 1095). It gives $(f_!\underline{\mathbf{Z}})^{\otimes p+1} = (W_p \to X)_!\underline{\mathbf{Z}} \oplus Rest$, which produces the comparison map towards the alternating power.
- The stalk computation identifying the quotient with the exterior power. Stalk computation over geometric points of X (proof lines 1103-1114). It identifies the kernel of the map to $(W_p \to X)_!\underline{\mathbf{Z}}$ with the tensors carrying a repeated factor, which is exactly the kernel of the map to the alternating power.

Source reference. Original lemma 0722: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 861–861. Navigation pointer for \ref{lemma-product-is-tensor-product} used inside the counted span (line 1095); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05W1: sources/stacks-chow/tags.txt, source lines 7570–7570. Navigation pointer for \ref{spaces-morphisms-lemma-diagonal-unramified-morphism} used inside the counted span (line 1090); the argument that consumes it is retained in the span.

Source reference. Original lemma 06CR: sources/stacks-chow/tags.txt, source lines 8155–8155. Navigation pointer for \ref{spaces-morphisms-lemma-etale-unramified} used inside the counted span (line 1089); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : U \to X$ be a surjective, étale, and separated morphism of algebraic spaces over S. For $p \geq 0$ set*

$$W_p = U \times_X \ldots \times_X U \setminus \textit{all diagonals}$$

where the fibre product has $p+1$ factors. There is a free action of S_{p+1} on W_p over X and

$$\mathrm{Hom}(K^p, \mathcal{F}) = S_{p+1}\textit{-anti-invariant elements of } \mathcal{F}(W_p)$$

functorially in $\mathcal{F}$ where $K^p = \wedge^{p+1} f_! \underline{\mathbf{Z}}$.

Proof. Because $U \to X$ is separated the diagonal $U \to U \times_X U$ is a closed immersion. Since $U \to X$ is étale the diagonal $U \to U \times_X U$ is an open immersion, see Morphisms of Spaces, Lemmas 06CR and 05W1. Hence W_p is an open and closed subspace of $U^{p+1} = U \times_X \ldots \times_X U$. The action of S_{p+1} on W_p is free as we've thrown out the fixed points of the action. By Lemma 0722 we see that

$$(f_! \underline{\mathbf{Z}})^{\otimes p+1} = f_!^{p+1} \underline{\mathbf{Z}} = (W_p \to X)_! \underline{\mathbf{Z}} \oplus \textit{Rest}$$

where $f^{p+1} : U^{p+1} \to X$ is the structure morphism. Looking at stalks over a geometric point $\overline{x}$ of X we see that

$$\left(\bigoplus\nolimits_{\overline{u} \mapsto \overline{x}} \mathbf{Z} \right)^{\otimes p+1} \longrightarrow (W_p \to X)_! \underline{\mathbf{Z}}_{\overline{x}}$$

is the quotient whose kernel is generated by all tensors $1_{\overline{u}_0} \otimes \ldots \otimes 1_{\overline{u}_p}$ where $\overline{u}_i = \overline{u}_j$ for some $i \neq j$. Thus the quotient map

$$(f_! \underline{\mathbf{Z}})^{\otimes p+1} \longrightarrow \wedge^{p+1} f_! \underline{\mathbf{Z}}$$

factors through $(W_p \to X)_! \underline{\mathbf{Z}}$, i.e., we get

$$(f_! \underline{\mathbf{Z}})^{\otimes p+1} \longrightarrow (W_p \to X)_! \underline{\mathbf{Z}} \longrightarrow \wedge^{p+1} f_! \underline{\mathbf{Z}}$$

This already proves that $\mathrm{Hom}(K^p, \mathcal{F})$ is (functorially) a subgroup of

$$\mathrm{Hom}((W_p \to X)_! \underline{\mathbf{Z}}, \mathcal{F}) = \mathcal{F}(W_p)$$

To identify it with the S_{p+1}-anti-invariants we have to prove that the surjection $(W_p \to X)_! \underline{\mathbf{Z}} \to \wedge^{p+1} f_! \underline{\mathbf{Z}}$ is the maximal S_{p+1}-anti-invariant quotient. In other words, we have to show that $\wedge^{p+1} f_! \underline{\mathbf{Z}}$ is the quotient of $(W_p \to X)_! \underline{\mathbf{Z}}$ by the subsheaf generated by the local sections $s - \mathrm{sign}(\sigma)\sigma(s)$ where s is a local section of $(W_p \to X)_! \underline{\mathbf{Z}}$. This can be checked on the stalks, where it is clear. □